AP BIOLOGY TOPIC 2.1

Cell Structure: Subcellular Components & Organelle Systems

UNIT 2: CELL STRUCTURE AND FUNCTION (10–13% EXAM WEIGHT)

SYI-1.D | SYI-1.E | ENDOMEMBRANE SYSTEM | ENERGY ORGANELLES

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 2.1 (CELL STRUCTURE: SUBCELLULAR COMPONENTS & ORGANELLE SYSTEMS)

CED Core Vocabulary

Ribosomes (Non-Membrane-Bound) — Composed of ribosomal RNA (rRNA) + proteins (two subunits: large & small; 70S in prokaryotes/mitochondria/chloroplasts vs. 80S in eukaryotic cytosol).

Free Ribosomes (suspended in cytosol) — Synthesize proteins that function within the cytosol (e.g., glycolysis enzymes, cytoskeletal proteins).

Nucleus & Nucleolus — Double-membrane nuclear envelope perforated by nuclear pores; houses linear DNA + histones (chromatin). Nucleolus synthesizes rRNA and assembles ribosomal subunits.

Lysosomes (Acidic Recycling Centers) — Membrane-bound sacs containing hydrolytic enzymes (acid hydrolases) that function optimally at acidic pH (~4.5–5.0) maintained by proton (H^+) pumps.

Function — Translate mRNA codons into polypeptide chains (protein synthesis).

Endomembrane System — Coordinated network of membranes (physically continuous or connected via transport vesicles) that synthesize, modify, package, and transport proteins &...

Golgi Complex (Apparatus) — Stack of flattened, polarized membranous sacs (cisternae): Cis face (receiving side): Near ER; receives transport vesicles from Rough ER.

Intracellular Digestion (Phagocytosis) — Fuse with food vacuoles/phagosomes to digest macromolecules & invading pathogens.

§2. UNIVERSAL CELL FEATURES & RIBOSOMES

Common Ancestry

- **Universal Features of All Cells (Prokaryotes & Eukaryotes):** Plasma membrane, cytosol (semifluid jelly), chromosomes (DNA), and **ribosomes** → strong evolutionary evidence of **common ancestry** (LUCA).
- **Ribosomes (Non-Membrane-Bound):**
  - Composed of **ribosomal RNA (rRNA) + proteins** (two subunits: large & small; **70S** in prokaryotes/mitochondria/chloroplasts vs. **80S** in eukaryotic cytosol).
  - **Function:** Translate mRNA codons into polypeptide chains (protein synthesis).
  - **Free Ribosomes (suspended in cytosol):** Synthesize proteins that function *within the cytosol* (e.g., glycolysis enzymes, cytoskeletal proteins).
  - **Bound Ribosomes (attached to Rough ER / nuclear envelope):** Synthesize proteins destined for *secretion (export), insertion into membranes, or lysosomes*.

[CORE AXIOM / REMEMBER] **Ribosomes = protein factories** of the cell. Because they lack a lipid membrane, they exist in every living organism across all three domains (Bacteria, Archaea, Eukarya).

[AP EXAM FRQ SCORING STRATEGY] If an AP question describes a cell secreting peptide hormones (e.g., insulin) or embedding receptor proteins into the plasma membrane → immediately think **Rough ER-bound ribosomes → Golgi → secretory vesicles**!

§5. LYSOSOMES, VACUOLES & CYTOSKELETON

Digestion & Support

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Lysosomes, Vacuoles & Cytoskeleton ➔ **Lysosomes (Acidic Recycling Centers)** ➔ **Cytoskeleton & Centrosomes**

- **Lysosomes (Acidic Recycling Centers):**
  - Membrane-bound sacs containing **hydrolytic enzymes (acid hydrolases)** that function optimally at **acidic pH (~4.5–5.0)** maintained by proton (**H^+**) pumps.
  - **Intracellular Digestion (Phagocytosis):** Fuse with food vacuoles/phagosomes to digest macromolecules & invading pathogens.
  - **Autophagy:** Recycle damaged/worn-out organelles into reusable monomers.
  - **Apoptosis (Programmed Cell Death):** Controlled release of hydrolytic/caspase enzymes sculptures digits during embryonic development or eliminates infected/cancerous cells.

[EXAMPLE & BIOLOGICAL APPLICATION] **Macrophages (white blood cells)** engulf bacteria via phagocytosis and fuse the phagosome with **lysosomes** to destroy the pathogen; tadpole tails are resorbed via lysosomal **apoptosis**.

VACUOLE TYPE	CELL TYPE & KEY FEATURES	PRIMARY BIOLOGICAL FUNCTIONS
Central Vacuole	**Plant cells**; single huge compartment (up to 80% of volume) bounded by *tonoplast*	Stores water, ions, pigments, & metabolic waste; generates **turgor pressure** against cell wall for plant support & cell growth
Food Vacuoles	**Animal cells & protists**; smaller, temporary, more numerous	Formed by phagocytosis; fuse with lysosomes for nutrient digestion & storage
Contractile Vacuole	**Freshwater protists** (e.g., *Paramecium*, *Amoeba*)	**Osmoregulation:** Actively pumps excess water out of cell to prevent osmotic lysis in hypotonic freshwater

- **Cytoskeleton & Centrosomes:** Dynamic network of protein fibers:
  - **Microtubules (tubulin dimers, Hollow, 25 nm):** Largest; maintain cell shape, form mitotic **spindle fibers** (grown from **centrosomes/centrioles**), tracks for motor proteins (kinesin/dynein), & core of **cilia/flagella** (9+2 arrangement).
  - **Microfilaments (actin filaments, 7 nm):** Thinnest; muscle contraction (with myosin), amoeboid pseudopodia, cytoplasmic streaming, and **cleavage furrow** in cytokinesis.
  - **Intermediate Filaments (keratin/lamins, 8–12 nm):** Permanent tension-bearing cables; anchor nucleus & organelles, form nuclear lamina.

§3. ENDOMEMBRANE SYSTEM: NUCLEUS, ER & GOLGI

Protein & Lipid Traffic

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Endomembrane System ➔ **Endomembrane System** ➔ **Nucleus & Nucleolus** ➔ **Golgi Complex (Apparatus)**

- **Endomembrane System:** Coordinated network of membranes (physically continuous or connected via **transport vesicles**) that synthesize, modify, package, and transport proteins & lipids.
  - **Includes:** Nuclear envelope, Rough & Smooth ER, Golgi apparatus, lysosomes, vacuoles, transport vesicles, and plasma membrane. *(Excludes mitochondria, chloroplasts, peroxisomes!)*
- **Nucleus & Nucleolus:** Double-membrane **nuclear envelope** perforated by nuclear pores; houses linear DNA + histones (chromatin). **Nucleolus** synthesizes rRNA and assembles ribosomal subunits.

ER TYPE	KEY STRUCTURE	MAIN FUNCTIONS
Rough ER (RER)	Studded with **bound ribosomes**; flattened cisternae continuous with nuclear envelope	**Synthesizes, folds & glycosylates** secretory/membrane/lysosomal proteins; packages them into transport vesicles; adds membrane phospholipids
Smooth ER (SER)	Branched tubular network; **NO ribosomes**	**Lipid & steroid hormone synthesis**, carbohydrate metabolism, **detoxification** of drugs/poisons (liver), **Ca^{2+}** storage (sarcoplasmic reticulum in muscle)

- **Golgi Complex (Apparatus):** Stack of flattened, polarized membranous sacs (**cisternae**):
  - ***Cis* face (receiving side):** Near ER; receives transport vesicles from Rough ER.
  - **Medial & *Trans* face (shipping side):** Modifies carbohydrate tags on glycoproteins (e.g., adds mannose-6-phosphate tag for lysosomes), folds/processes molecules, sorts, and packages them into vesicles for secretion or delivery.

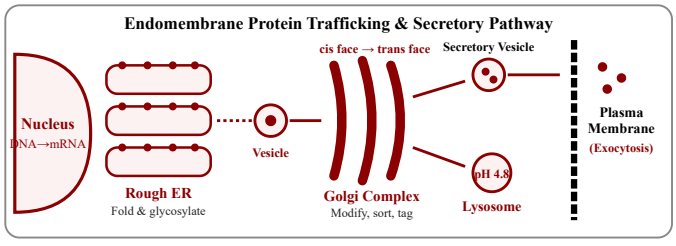
[AP EXAM FRQ SCORING STRATEGY] **Rough ER → proteins** (secretion/membrane) | **Smooth ER → lipids + detoxification + Ca^{2+}** | **Golgi = post office / packaging & sorting center** (*cis* receives → *trans* ships).

[MECHANISTIC NOTE] **CED Scope Boundary:** Specific functions of Smooth ER in specialized cells and specific Golgi enzymes for synthesizing particular phospholipids or packaging peroxisomal vs. secretory enzymes are beyond the AP Exam scope.

§4. SECRETORY PATHWAY VISUALIZED (DNA → EXOCYTOSIS)

High-Yield Pathway

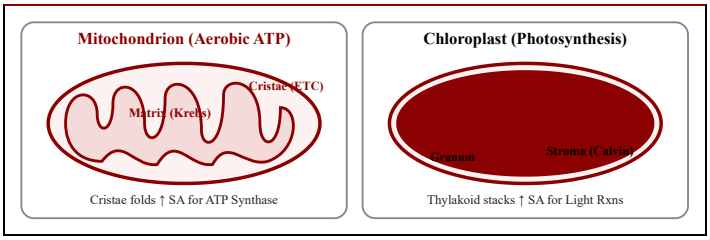
1. Nucleus (mRNA) → **2. Bound Ribosome / RER** → **3. Transport Vesicle** → **4. Cis → Trans Golgi** → **5. Secretory Vesicle → Plasma Membrane**



§6. MITOCHONDRIA & CHLOROPLASTS (ENERGY ORGANELLES)

Double Membrane + DNA

- **Mitochondria (Site of Aerobic Cellular Respiration):**
  - Found in **nearly ALL eukaryotic cells** (including plants, animals, fungi, protists!).
  - **Outer Membrane:** Smooth, porous (contains porin proteins).
  - **Inner Membrane (Cristae):** Highly folded infoldings called **cristae** that dramatically **increase surface area** to embed more Electron Transport Chain (ETC) proteins & ATP synthase complexes → maximizes ATP synthesis.
  - **Compartments: Intermembrane space** (narrow region where **H^+** accumulates to low pH) and **Mitochondrial Matrix** (contains Krebs/citric acid cycle enzymes, mitochondrial circular DNA, & 70S ribosomes).
- **Chloroplasts (Site of Photosynthesis in Plants & Algae):**
  - Bounded by a smooth **outer membrane** and **inner membrane** separated by a narrow intermembrane space.
  - **Thylakoids & Grana:** Internal system of flattened membranous sacs (**thylakoids**) stacked into **grana**; thylakoid membranes house chlorophyll, photosystems I & II, and ETC for the **light-dependent reactions** (pumping **H^+** into the **thylakoid lumen**).
  - **Stroma:** Fluid outside thylakoids containing Rubisco & enzymes for the **Calvin cycle (carbon fixation)**, circular chloroplast DNA, and 70S ribosomes.



[MECHANISTIC NOTE] **Complementary Energy Organelles: Chloroplasts** capture solar light energy to synthesize organic sugars (**$CO_2 + H_2O \rightarrow C_6H_{12}O_6 + O_2$**); **Mitochondria** oxidize organic sugars to release energy as ATP (**$C_6H_{12}O_6 + O_2 \rightarrow CO_2 + H_2O + ATP$**). *Plant cells have BOTH!*

[CORE AXIOM / REMEMBER] **Putting It All Together (Coordinated System):**

- **Ribosome** → polypeptide synthesis | **Rough ER** → protein folding & transport
- **Smooth ER** → lipid synthesis & detox | **Golgi** → modify, package, & ship
- **Mitochondria** → aerobic ATP production | **Chloroplast** → photosynthesis
- **Lysosome** → acidic digestion, autophagy, & apoptosis | **Vacuole** → storage & turgor

§7. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 2.1 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 2.1: Cell Structure: Subcellular Components & Organelle Systems ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 4 (Diffusion & Osmosis): Dialysis tubing & potato core mass change (% ΔMass = [(Final − Initial) / Initial] × 100) across sucrose molarities (0.0 M to 1.0 M). IV: Extracellular sucrose molarity (M); DV: % change in mass; Isotonic point: x-intercept where % ΔMass = 0% (no net water movement). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Graph Rules: Total Water Potential: $\Psi = \Psi_p + \Psi_s$. Solute Potential: $\Psi_s = -iCRT$ (where i = ionization constant [1 for sucrose/glucose, 2 for NaCl], C = molarity in mol/L, R = 0.0831 L·bar/mol·K, T = K = °C + 273). Surface-Area-to-Volume Ratio (SA:V): Cube SA = $6s^2$, V = s^3 (SA:V = 6/s). Larger SA:V accelerates diffusion per unit volume. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Physiological Disruption Case Study: Cystic Fibrosis results from a deletion (ΔF508) in the CFTR chloride channel protein, preventing misfolded channels from reaching the apical plasma membrane. Impaired Cl^- efflux prevents osmotic water movement into mucus, producing thick, dehydrated mucus in lungs and pancreatic ducts. Ouabain/Digoxin inhibits the Na^+/K^+-ATPase pump, collapsing the electrochemical gradient required for secondary active cotransport.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 2.1 to prove why the claim holds.

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